# Cleaning medical data

Code

FAROUK OSMAN

2026-09-02

## 1 Introduction

This is a data cleaning excercise on a raw dataset that was posted in Kaggle;
https://www.kaggle.com/datasets/aamir5659/raw-medical-dataset-for-cleaning-practice?resource=download

The dataset contains the following variables:

- **Patient_ID** Unique patient identifier
- **Age** Patient age in years(numeric variable)
- **Gender** Patient gender(Categorical; Male/Female)
- **Blood_Pressure** Systolic blood pressure in mmHg(numeric)
- **Cholesterol** Total cholesterol in mg/dL(numeric)
- **BMI** Body Mass Index(numeric)
- **Smoker** Smoking status (Categorical; Yes/No)
- **Diagnosis** Medical condition diagnosed(Categorical; Heart disease,Diabetes and None)
- **Admission_Date** Date of hospital admission(date)
- **Notes** Free text clinical notes(character)

### 1.1 Libraries

Hide

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

Hide

```
library(janitor)
```

```
##
## Attaching package: 'janitor'
```

```
## The following objects are masked from 'package:stats':
##
```

```
##     chisq.test, fisher.test
```

Hide

```
library(skimr)
library(here)
```

```
## here() starts at E:/Documents/R/cleaning medical data
```

## 1.2 Loading the Data

Hide

```
medical <- read.csv(here("data/realworld_medical_dirty.csv"))
```

# 2 Inspection and Data Audit

## 2.1 Structure and Dimensions

Hide

```
glimpse(medical)
```

```
## Rows: 100
## Columns: 10
## $ Patient_ID     <chr> "P1000", "P1001", "P1002", "P1003", "P1004", "P1005", "…
## $ Age            <dbl> 55, 65, 45, 65, 65, 35, 45, 45, 45, 65, 55, 45, NA, 65,…
## $ Gender         <chr> "FEMALE", "Male", "Male", "Male", "Male", "Female", "MA…
## $ Blood_Pressure <dbl> 120, 150, 120, NA, 120, 130, 150, 150, NA, 130, 140, 12…
## $ Cholesterol    <dbl> NA, 180, 220, 180, 300, 200, NA, 200, 200, NA, 220, 300…
## $ BMI            <dbl> 35.4, 27.8, 22.5, 35.4, 40.1, 35.4, 22.5, NA, NA, 35.4,…
## $ Smoker         <chr> "Yes", "nan", "No", "No", "nan", "N", "No", "N", "No", …
## $ Diagnosis      <chr> "Heart Disease", "None", "nan", "nan", "None", "Heart D…
## $ Admission_Date <chr> "2023-12-27", "2023-01-01", "2023-12-14", "2023-07-09",…
## $ Notes          <chr> "", "", "", "", "Follow-up required", "Follow-up requir…
```

This data set contains 7 variables, 1 column for the Patient ID, Admission dates and Clinical notes with 100 patients. There are multiple missing values and these will be investigated in several columns. Gender and Smoker appear to have multiple iterations of Male/Female or Yes/No

## 2.2 Assess missingness and Plausability

Hide

```
skim(medical)
```

| | |
|---|---|
| Name | medical |
| Number of rows | 100 |
| Number of columns | 10 |

_____

| | |
|---|---|
| Column type frequency: | |
| character | 6 |
| numeric | 4 |
| ——————————— | |
| Group variables | None |

Data summary

**Variable type: character**

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---|---|---|---|---|---|---|---|
| Patient_ID | 0 | 1 | 5 | 5 | 0 | 100 | 0 |
| Gender | 0 | 1 | 3 | 6 | 0 | 6 | 0 |
| Smoker | 0 | 1 | 1 | 3 | 0 | 5 | 0 |
| Diagnosis | 0 | 1 | 3 | 13 | 0 | 4 | 0 |
| Admission_Date | 0 | 1 | 10 | 10 | 0 | 88 | 0 |
| Notes | 0 | 1 | 0 | 18 | 23 | 5 | 0 |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| Age | 17 | 0.83 | 47.17 | 13.62 | 25.0 | 35.0 | 55.0 | 55.00 | 65.0 | ▅▃▆▇ |
| Blood_Pressure | 14 | 0.86 | 137.09 | 13.36 | 120.0 | 120.0 | 140.0 | 150.00 | 160.0 | ▇▁▇▃ |
| Cholesterol | 20 | 0.80 | 226.75 | 43.22 | 180.0 | 195.0 | 220.0 | 250.00 | 300.0 | ▇▃▁▂ |
| BMI | 20 | 0.80 | 32.23 | 6.24 | 22.5 | 27.8 | 35.4 | 36.57 | 40.1 | ▅▂▇▇ |

For numeric variables; There are ;

- 17 missing values in the Age variable
- 14 missing values in the Blood_Pressure
- 20 missing values in the Cholesterol column
- 20 missing values in the BMI column

All numeric variables are within clinicaly plausible range

## 2.3 Identify Fake “nan” Strings

Hide

```
# Count how many cells contain the string "nan" in each column
colSums(medical == "nan", na.rm = TRUE)
```

```
##     Patient_ID            Age         Gender Blood_Pressure    Cholesterol
##              0              0             15              0              0
##            BMI         Smoker      Diagnosis Admission_Date          Notes
##              0             20             21              0              0
```

Results showed:

- Gender: 15
- Smoker: 20
- Diagnosis: 21

## 2.4 Distribution of the categorical variables

Gender

Hide

```
#Gender
tabyl(medical$Gender)
```

| medical$Gender | n | percent |
|---|---|---|
| <chr> | <int> | <dbl> |
| FEMALE | 15 | 0.15 |
| Female | 17 | 0.17 |
| MALE | 21 | 0.21 |
| Male | 22 | 0.22 |
| male | 10 | 0.10 |
| nan | 15 | 0.15 |

6 rows

Gender should be distributed as follows; Male, Female or NA for missing values; - “FEMALE” should be Female - “MALE” should be Male - “male” should be Male - “nan” should be NA

Smoker

Hide

```
tabyl(medical$Smoker)
```

| medical$Smoker | n | percent |
|---|---|---|
| <chr> | <int> | <dbl> |
| N | 20 | 0.20 |
| No | 20 | 0.20 |
| Y | 17 | 0.17 |
| Yes | 23 | 0.23 |
| nan | 20 | 0.20 |

5 rows

Smoker should be “Yes/No” - “N” should be “No” - “Y” should be “Yes” - “nan” should be “NA”

Diagnosis

Hide

```
tabyl(medical$Diagnosis)
```

| medical$Diagnosis | n | percent |
|---|---|---|
| <chr> | <int> | <dbl> |
| Diabetes | 27 | 0.27 |
| Heart Disease | 25 | 0.25 |
| None | 27 | 0.27 |
| nan | 21 | 0.21 |

4 rows

Diagnosis should have Diabetes, Heart Disease and None - “nan” should be NA

# 3 Cleaning

We have to address the following in this order

1. remove the fake “nan” strings and convert to a missing value
2. Correct the distribution of categorical variables
3. Convert the empty strings in notes to NA
4. Impute missing numeric variables using median
5. Convert admission date to real dates

## 3.1 Remove “nan” to NA

Hide

```
medical$Gender[medical$Gender == "nan"] <- NA
medical$Smoker[medical$Smoker == "nan"] <- NA
medical$Diagnosis[medical$Diagnosis == "nan"] <- NA
```

Check

Hide

```
colSums(medical == "nan", na.rm = TRUE )
```

```
##     Patient_ID            Age         Gender Blood_Pressure    Cholesterol 
##              0              0              0              0              0 
##            BMI         Smoker      Diagnosis Admission_Date          Notes 
##              0              0              0              0              0
```

All “nan” strings removed

## 3.2 Correct distribution of categorical variables

Gender

Hide

```
medical$Gender[medical$Gender == "FEMALE"] <- "Female"
medical$Gender[medical$Gender == "MALE"] <- "Male"
medical$Gender[medical$Gender == "male"] <- "Male"
tabyl(medical$Gender)
```

| | medical$Gender | n | percent | valid_percent |
|---|---|---|---|---|
| | <chr> | <int> | <dbl> | <dbl> |
| 1 | Female | 32 | 0.32 | 0.3764706 |
| 2 | Male | 53 | 0.53 | 0.6235294 |
| 3 | *NA* | 15 | 0.15 | *NA* |

3 rows

Smoker

Hide

```
medical$Smoker[medical$Smoker == "N"] <- "No"
medical$Smoker[medical$Smoker == "Y"] <- "Yes"
tabyl(medical$Smoker)
```

| | medical$Smoker | n | percent | valid_percent |
|---|---|---|---|---|
| | <chr> | <int> | <dbl> | <dbl> |
| 1 | No | 40 | 0.4 | 0.5 |
| 2 | Yes | 40 | 0.4 | 0.5 |
| 3 | *NA* | 20 | 0.2 | *NA* |

3 rows

Diagnosis

Hide

```
tabyl(medical$Diagnosis)
```

| | medical$Diagnosis | n | percent | valid_percent |
|---|---|---|---|---|
| | <chr> | <int> | <dbl> | <dbl> |
| 1 | Diabetes | 27 | 0.27 | 0.3417722 |
| 2 | Heart Disease | 25 | 0.25 | 0.3164557 |
| 3 | None | 27 | 0.27 | 0.3417722 |
| 4 | *NA* | 21 | 0.21 | *NA* |

4 rows

## 3.3 Convert the empty strings and N/A in notes to NA

Hide

```
medical$Notes[medical$Notes == ""] <- NA
medical$Notes[medical$Notes == "N/A"] <- NA
tabyl(medical$Notes)
```

| | medical$Notes | n | percent | valid_percent |
|---|---|---|---|---|
| | <chr> | <int> | <dbl> | <dbl> |
| 1 | Checkup due | 18 | 0.18 | 0.3157895 |
| 2 | Follow-up required | 25 | 0.25 | 0.4385965 |
| 3 | Pending | 14 | 0.14 | 0.2456140 |
| 4 | *NA* | 43 | 0.43 | *NA* |

4 rows

## 3.4 Impute missing numeric variables using median

Imputing shall be done for; Age, Blood pressure, Cholesterol and BMI

Hide

```
medical$Age[is.na(medical$Age)] <- median(medical$Age, na.rm = TRUE)
medical$Blood_Pressure[is.na(medical$Blood_Pressure)] <- median(medical$Blood_Pressure, na.
medical$Cholesterol[is.na(medical$Cholesterol)] <- median(medical$Cholesterol, na.rm = TRUE
medical$BMI[is.na(medical$BMI)] <- median(medical$BMI, na.rm = TRUE)
```

Check

Hide

```
colSums(is.na(medical))
```

```
##     Patient_ID            Age         Gender Blood_Pressure    Cholesterol
##              0              0             15              0              0
##            BMI         Smoker      Diagnosis Admission_Date          Notes
##              0             20             21              0             43
```

None of the numeric variables now have missing values

## 3.5 Convert Admission date to date format

Hide

```
medical <-  medical |>
  mutate(
    Admission_Date = as.Date(Admission_Date, format = "%Y-%m-%d")
  )
class(medical$Admission_Date)
```

```
## [1] "Date"
```

## 3.6 Convert the categorical variables to factor

Hide

```
medical <- medical |>
  mutate(
    Gender = as.factor(Gender),
    Smoker = as.factor(Smoker),
    Diagnosis = as.factor(Diagnosis),
    Notes     = as.character(Notes)
  )
glimpse(medical)
```

```
## Rows: 100
## Columns: 10
## $ Patient_ID     <chr> "P1000", "P1001", "P1002", "P1003", "P1004", "P1005", "…
## $ Age            <dbl> 55, 65, 45, 65, 65, 35, 45, 45, 45, 65, 55, 45, 55, 65,…
## $ Gender         <fct> Female, Male, Male, Male, Male, Female, Male, Male, Mal…
## $ Blood_Pressure <dbl> 120, 150, 120, 140, 120, 130, 150, 150, 140, 130, 140, …
```

```
## $ Cholesterol    <dbl> 220, 180, 220, 180, 300, 200, 220, 200, 200, 220, 220, …
## $ BMI            <dbl> 35.4, 27.8, 22.5, 35.4, 40.1, 35.4, 22.5, 35.4, 35.4, 3…
## $ Smoker         <fct> Yes, NA, No, No, NA, No, No, No, No, No, Yes, No, Yes, …
## $ Diagnosis      <fct> Heart Disease, None, NA, NA, None, Heart Disease, None,…
## $ Admission_Date <date> 2023-12-27, 2023-01-01, 2023-12-14, 2023-07-09, 2023-0…
## $ Notes          <chr> NA, NA, NA, NA, "Follow-up required", "Follow-up requir…
```

As noted all variables are now in their appropriate classes.

## 3.7 Export the cleaned file

Hide

```
write.csv(medical, here("data/medical_clean.csv"))
```